

Polarized SI-Membrane Cosmology, Opus III: Universal Matter Orientation, Branch Condensation, and Relic Statistics

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July 5, 2026

Corrected draft manuscript

DOI: 10.5281/zenodo.21312037

Abstract

Opus I introduced polarized membrane cosmology as a three-sector yield problem: a higher-dimensional pre-projection structure produces a visible matter-oriented branch A , a conjugate branch B , and a neutral relic sector R associated with the branch-separating order parameter. Opus II sharpened this picture by naming the membrane SI and by proposing that the relevant sorting variable is a universal orientation label, not an ordinary Standard Model gauge charge.

Opus III corrects and extends that framework. The central claim is that matter-antimatter orientation is not generated independently for each particle species. Rather, it is a pre-particle label carried by a deeper higher-dimensional excitation. Observed particles are quantized projection modes of that excitation. The apparent coordination between electron, quark, baryon, and lepton asymmetries is therefore not treated as a late bookkeeping miracle, but as evidence that one universal orientation projects into many Standard Model channels.

This note also corrects three possible misunderstandings. First, SI is not a local positron-exclusion rule: local QED, pair production, and annihilation remain valid inside branch A . Second, the scale near 40 MeV is not a static particle barrier. It is the approximate temperature at which the SI order parameter condenses and the universal orientation label becomes branch-classical during the annihilation/projection history. Third, the relic sector R is not a third observer-branch equivalent to A and B . It is neutral residual stress-energy of the SI transition, potentially accessible to branch A gravitationally while remaining neutral under Standard Model gauge interactions.

The dark matter target is recast as a relic-statistics problem. Since observations fix the dark-to-baryonic density ratio near 5.3, a successful SI model must predict the relic-to-baryon number ratio. Once n_R/n_b is known, the relic mass is fixed by $m_R/m_p \simeq 5.3(n_b/n_R)$. Thus the model cannot merely invoke “topological relics.” It must derive their statistics from SI geometry, projection multiplicities, or index structure.

1 Purpose of Opus III

The observed universe is often summarized as matter-dominated. This is true, but incomplete.

We do not merely observe that baryons outnumber antibaryons in our visible universe. We also observe that the surviving charged matter is coordinated across sectors. The excess of negative leptonic charge matches the excess of positive baryonic charge well enough that the universe is electrically neutral on large scales.

This does not mean that standard physics is incapable of charge bookkeeping. Standard thermal physics, chemical equilibrium, gauge neutrality, and reaction networks can redistribute a given asymmetry across available particle species. If an asymmetry such as $B - L$ is already present, known plasma reactions can spread it among quarks, leptons, and other degrees of freedom.

Opus III does not deny this.

The question asked here is different:

What is the asymmetry before particle species are already separated into electrons, quarks, neutrinos, baryons, and dark-sector remnants?

The proposal of Opus III is that the matter-antimatter distinction is not first a property of individual particles. It is a universal pre-particle orientation label of a deeper excitation. Particles are the discrete projection modes through which that label becomes visible.

In compact form: one does not first generate many synchronized asymmetries. One begins with one universal orientation asymmetry, and the observed particle asymmetries are its projected components.

This is the central inversion of Opus III.

2 What standard physics explains, and what it does not explain here

Standard cosmology already contains powerful charge-accounting mechanisms. A reaction such as

$$a + b \leftrightarrow c + d \tag{1}$$

imposes chemical-potential relations in equilibrium:

$$\mu_a + \mu_b = \mu_c + \mu_d. \tag{2}$$

Gauge neutrality imposes constraints such as

$$\sum_i q_i (n_i - n_{\bar{i}}) = 0. \tag{3}$$

Therefore, once an asymmetry is present, the early plasma can redistribute it. There is no need for a literal cosmic accountant making electrons “catch up” with protons one by one.

However, this does not answer the deeper question posed by the SI program. The SI question is not:

How does the plasma redistribute an already-existing asymmetry?

The SI question is:

Why is there a universal matter-antimatter orientation capable of being redistributed across the particle spectrum in the first place?

In ordinary language: standard physics can explain how the imbalance is shared among particles. Opus III asks what the imbalance is before “particle type” is the primary description.

Thus the claim is not that charge neutrality is impossible in standard cosmology. The claim is that the observed universality of matter orientation is suggestive of a deeper label that is projected into the Standard Model spectrum.

3 The universal matter label

Let the deeper excitation be denoted schematically as

$$|X, \chi, \lambda\rangle, \quad (4)$$

where X is the underlying higher-dimensional excitation, χ is the universal matter-antimatter orientation, and λ is a projection mode that becomes a particle state after membrane projection.

The two possible orientations are

$$\chi = +, \quad \chi = -. \quad (5)$$

These are not yet “electron” and “positron,” nor “quark” and “antiquark.” They are prior to the resolved particle spectrum.

The observed particle types arise from projection:

$$\mathcal{P}_A |X, \chi = +, \lambda_i\rangle \mapsto |f_i\rangle_A, \quad (6)$$

$$\mathcal{P}_B |X, \chi = -, \lambda_i\rangle \mapsto |\bar{f}_i\rangle_B, \quad (7)$$

where f_i may appear as electron, muon, tau, quark, neutrino, or another projected mode depending on λ_i .

The important point is this: the particle zoo is secondary. The universal orientation label is primary.

Electrons and quarks do not need to independently acquire matching asymmetries after the fact. Their asymmetries are correlated because they are different projection modes of the same oriented parent structure. Thus

$$\Delta n_i = c_i \Delta \chi, \quad (8)$$

where c_i is determined by projection structure, reaction history, gauge constraints, and survival through the thermal plasma.

The coefficients c_i are not arbitrary fitting constants in a completed theory. They must eventually be derived from the SI projection operator, Standard Model reaction network, and the timing of the SI transition.

4 The three-sector yield, clarified

Opus I wrote the schematic three-sector yield

$$\mathcal{H}_{\text{deep}} \longrightarrow \mathcal{H}_A \oplus \mathcal{H}_B \oplus \mathcal{H}_R. \quad (9)$$

Opus III keeps this architecture, but clarifies the meaning of the sectors.

A and B are conjugate branch sectors. A is the visible branch in which the $\chi = +$ orientation becomes the matter-oriented background. B is the conjugate branch associated with $\chi = -$.

R is not a third observer branch equivalent to A and B . It is the neutral residual sector of the SI transition: compact excitations, hidden order-parameter remnants, or stable projection residues associated with incomplete condensation of the branch-separating structure.

Thus the more precise reading is:

$\mathcal{H}_{\text{deep}}$ produces an A -branch yield, a B -branch yield, and a neutral SI relic yield.

The R sector is part of the post-transition stress-energy output of the projection event. It is not ordinary antimatter hidden in another spatial region, and it is not a third copy of the visible universe.

For observers in A , the relevant stress-energy can include

$$\rho_{\text{obs},A} = \rho_{b,A} + \rho_{\gamma,A} + \rho_{R,A} + \cdots, \quad (10)$$

where $\rho_{R,A}$ denotes the relic component gravitationally accessible inside branch A .

The conjugate branch B is not assumed to be electromagnetically, thermally, or directly gravitationally visible inside A . If there is any common channel, it must be specified by the future theory. The minimal working assumption is that B is branch-isolated, while the A -accessible relic component R_A is produced as neutral residue of the same SI transition.

This avoids treating R as “another universe” while still preserving the original three-sector yield idea.

5 Gauge neutrality is mandatory

The SI membrane cannot sort isolated Standard Model charges into different macroscopic sectors. A universe with leftover protons but no corresponding electron-sector excess would be electrically charged. That is not our universe. Annihilation cannot repair this mismatch because annihilation removes symmetric pairs; it preserves net charge differences.

Therefore the SI projection must act on gauge-compatible collective sectors, not on isolated particles one by one. The parent projection state must obey something like

$$Q_{\text{deep}} = 0, \quad (11)$$

and the branch-visible projection must preserve

$$Q_A = 0, \quad Q_B = 0, \quad Q_{R,\text{SM}} = 0. \quad (12)$$

This does not mean every projected particle is neutral. It means the collective projected sector is gauge-neutral.

The branch A projection may contain protons, electrons, quarks, neutrinos, photons, and other excitations, but the long-range Standard Model gauge charges must cancel macroscopically.

Thus Opus III should not say: protons are sent to A , antiprotons are sent to B , and electrons somehow catch up later. It should say: a charge-neutral matter-oriented sector is projected into A . Likewise, the conjugate charge-neutral sector is projected into B .

In compact form:

$$\mathcal{P}_A|X, \chi = +, \Lambda; Q = 0\rangle \rightarrow \text{matter-oriented plasma in } A, \quad Q_A = 0, \quad (13)$$

$$\mathcal{P}_B|X, \chi = -, \Lambda; Q = 0\rangle \rightarrow \text{conjugate-oriented plasma in } B, \quad Q_B = 0. \quad (14)$$

The charge constraint belongs to the parent sector. The particle content is its projected expression.

6 Local antiparticles remain allowed

The SI projection is not a local antiparticle ban. Local quantum field theory must remain valid inside branch A . In particular,

$$\gamma + \gamma \leftrightarrow e^- + e^+ \quad (15)$$

must remain allowed.

Particle accelerators, cosmic-ray interactions, beta processes, pair creation, and ordinary annihilation all occur within branch A . These are branch-internal processes. They do not require crossing the primordial SI projection boundary.

Therefore the following interpretation is forbidden:

e^- belongs to A , while e^+ is projected out of A .

That would contradict observed physics.

The correct statement is: the SI transition selects the large-scale branch orientation of the residual cosmological background. It does not remove local antiparticle excitations from the effective field theory of branch A .

A positron produced in branch A is an excitation of branch- A fields. It is not a piece of branch B leaking back into our universe. This is essential. Without it, the theory would immediately fail local QED.

7 The SI transition is not a particle-energy barrier

Opus II described the SI threshold in language that could be read as a static particle barrier:

$$E \gg E_{\text{SI}} \Rightarrow \text{free crossing}, \quad (16)$$

$$E \lesssim E_{\text{SI}} \Rightarrow \text{blocked crossing}. \quad (17)$$

Read literally, this is dangerous. If E_{SI} were simply around 40 MeV, then modern high-energy cosmic rays or collider particles would have energies far above the threshold and should leak across the membrane. That is not observed.

Opus III corrects the interpretation. The 40 MeV scale is not the height of a present-day particle barrier. It is the approximate critical temperature of the SI order-parameter transition.

Above T_c , the SI order parameter is uncondensed or branch-coherent:

$$\langle \phi_{\text{SI}} \rangle = 0. \quad (18)$$

Below T_c , the order parameter condenses:

$$\langle \phi_{\text{SI}} \rangle = \pm v_0. \quad (19)$$

After condensation, branch orientation becomes dynamically classical and cross-branch amplitudes are suppressed by the post-transition structure. A modern TeV particle does not cross merely because TeV is larger than 40 MeV. The relevant object is not a static 40 MeV wall. It is a condensed branch-separating phase.

The slogan is:

The particles do not cool below the membrane. The membrane freezes around the particles.

8 Why T_c around 40 MeV still matters

The SI orientation label χ is not created at T_c . It is deeper than the thermal plasma. However, T_c still matters because it determines when χ becomes a branch-level occupation asymmetry during the annihilation and projection history.

If SI condensation happens too early, the model risks isolating sectors before the ordinary annihilation history has produced the observed photon bath. If SI condensation happens too late, ordinary annihilation may erase the relevant branch residue below the observed level, or the projection may no longer explain why the visible residue has its observed scale.

Therefore T_c does not create χ , but it controls when χ becomes branch-classical. The visible baryon residue should not be written as a free upstream asymmetry alone,

$$\eta_A \neq \text{arbitrary } \Delta\chi. \quad (20)$$

Nor should it be written as pure thermal annihilation without projection. The intended structure is

$$\eta_A = F(\Delta\chi, T_c, \Gamma_{\text{ann}}, H, S_{\text{SI}}, \text{projection spectrum}). \quad (21)$$

Here $\Delta\chi$ is the deeper orientation imbalance or occupation bias; T_c is the SI condensation temperature; Γ_{ann} is the annihilation/reaction rate history; H is the expansion rate; S_{SI} represents the branch-separating action or condensation dynamics; and the projection spectrum determines how χ appears as baryonic, leptonic, radiative, and relic output.

Thus 40 MeV is not decorative. It is the moment at which the universal orientation label becomes locked into branch populations while annihilation history is still physically relevant.

A useful Opus IV calculation must show that this timing produces a visible residue near

$$\eta_B \approx 6 \times 10^{-10} \quad (22)$$

without fitting it independently from the relic abundance.

9 Neutral and self-conjugate fields

The matter-orientation label χ acts most naturally on oriented matter sectors. But not every field has an obvious matter-antimatter distinction. Photons are self-conjugate. The neutral pion is self-conjugate. The Z boson is self-conjugate. If neutrinos are Majorana particles, then neutrino and antineutrino are not distinct in the same way as electron and positron.

Therefore Opus III must not pretend that every degree of freedom carries χ in the same way. The proposed treatment is:

Self-conjugate radiation is not sorted by χ as matter or antimatter. It becomes part of the branch-internal environment after projection.

Photons belong to branch A once branch A exists, but they are not matter-oriented in the same sense as fermionic matter. The photon bath is therefore not duplicated by χ sorting, nor is it thrown

into the relic sector. It is part of the thermal branch environment produced by ordinary annihilation and plasma evolution.

Neutrinos remain an exposure point. The minimal version of the SI orientation picture is most naturally compatible with Dirac-like neutrino bookkeeping, where neutrino and antineutrino orientation can be tracked. If neutrinos are Majorana, then the χ map must be reformulated for the neutral lepton sector.

A confirmed Majorana neutrino would not automatically destroy every possible higher-dimensional orientation theory, but it would falsify the minimal Dirac-style version unless a Majorana-compatible projection rule is supplied. Thus the neutrino sector is a real test, not a decorative aside.

10 Relic statistics, not vague relics

The relic sector R cannot be invoked vaguely. The observed dark matter density is already known relative to baryonic matter:

$$\frac{\rho_{\text{DM}}}{\rho_b} \approx 5.3. \quad (23)$$

If the SI relic sector is the dark matter, then

$$\frac{\rho_R}{\rho_b} \approx 5.3. \quad (24)$$

But

$$\rho_R = m_R n_R, \quad (25)$$

and approximately

$$\rho_b = m_p n_b. \quad (26)$$

Therefore

$$m_R n_R \approx 5.3 m_p n_b. \quad (27)$$

Thus

$$\boxed{\frac{m_R}{m_p} \approx 5.3 \left(\frac{n_b}{n_R} \right)}. \quad (28)$$

This is the key relic-statistics equation. The model cannot choose m_R freely once n_R is known. Conversely, the model cannot choose n_R freely once m_R is known. A predictive SI theory must derive the relic-to-baryon number ratio

$$\kappa_R = \frac{n_R}{n_b}. \quad (29)$$

Then

$$m_R \approx \frac{5.3}{\kappa_R} m_p. \quad (30)$$

Several limiting cases illustrate the constraint. If $n_R \approx n_b$, then m_R is around 5 GeV; this resembles asymmetric dark matter accounting. If n_R is much larger than n_b , the relic mass must be much smaller. If n_R is much smaller than n_b , the relic mass must be much larger.

If one assumes one relic per horizon patch at T_c , then relics are extremely rare and must be enormously massive. That is not a generic requirement of SI theory; it is only one possible Kibble-style statistic. Opus III should not assume it.

Instead, the SI program must derive n_R from projection statistics, index multiplicities, or compact

relic formation dynamics.

The correct dark-matter question is therefore:

What relic number density does the SI transition produce?

Not merely:

Can SI produce some relics?

11 Relation to asymmetric dark matter

The relic-statistics equation makes contact with asymmetric dark matter. Asymmetric dark matter models explain the similarity between dark and baryonic densities by tying visible and dark number asymmetries to a common origin.

Opus III should acknowledge this contact. However, the SI proposal is not meant to be ordinary asymmetric dark matter with new notation. In conventional asymmetric dark matter, one typically begins with already-defined visible and dark sectors, then introduces transfer interactions that relate their asymmetries.

In Opus III, the claim is deeper: the visible matter sector, conjugate branch sector, and relic sector are all projection outputs of a pre-particle SI structure. Thus the shared asymmetry is not merely transferred between completed sectors. It is inherited from a common parent orientation.

The accounting may resemble asymmetric dark matter. The proposed origin is different.

The SI program becomes nontrivial only if it derives one of the quantities ordinary ADM models insert phenomenologically, especially

$$\kappa_R = \frac{n_R}{n_b}, \quad (31)$$

or an equivalent index quotient. If κ_R remains symbolic, the model has not yet advanced beyond bookkeeping.

12 Density-matrix form and branch condensation

Before SI condensation, the deep state need not be diagonal in the branch orientation basis. A schematic branch-orientation density matrix may be written as

$$\rho_\chi = \begin{pmatrix} AA & AB \\ BA & BB \end{pmatrix}. \quad (32)$$

Above T_c , off-diagonal terms may be present:

$$AB, BA \neq 0. \quad (33)$$

Below T_c , the SI order parameter selects a pointer basis, and the off-diagonal terms are damped:

$$AB, BA \rightarrow 0. \quad (34)$$

A schematic evolution is

$$\dot{\rho} = -i[H, \rho] - \Gamma_{\text{SI}}(T)[M_{\text{SI}}, [M_{\text{SI}}, \rho]] + \cdots, \quad (35)$$

where

$$\Gamma_{\text{Sl}}(T) \approx 0 \quad (T > T_c), \quad (36)$$

$$\Gamma_{\text{Sl}}(T) \neq 0 \quad (T < T_c). \quad (37)$$

This language should not be confused with an ordinary spatial wall separating matter and antimatter domains inside our observable universe. The Sl transition is not a patchwork of matter and antimatter regions in the same three-dimensional space. Such a model would face annihilation-front constraints.

Instead, Sl condensation selects branch sectors of the projected state. Ordinary Standard Model observables in branch A have negligible matrix elements connecting them to branch B .

The relic sector R is not branch B . It is neutral residual stress-energy of the Sl transition that may remain in the effective branch- A stress-energy budget.

13 What Opus III claims

Opus III claims:

1. Matter-antimatter orientation is a universal pre-particle label.
2. Observed particles are quantized projection modes of a deeper excitation.
3. The apparent coordination of charged-sector asymmetries reflects one projected orientation structure, not many independent accidents.
4. Local antiparticles remain allowed inside branch A .
5. Gauge neutrality is mandatory at the level of projected collective sectors.
6. The Sl transition near $T_c \approx 40 \text{ MeV}$ is a condensation event, not a static particle barrier.
7. T_c matters because it locks the orientation label into branch populations during the annihilation/projection history.
8. The relic sector R is neutral Sl transition residue, not an ordinary antimatter domain and not a third observer branch equivalent to B .
9. Dark matter must be treated through relic statistics: $m_R n_R = \rho_{\text{DM}}$.
10. The model becomes predictive only if it derives η_B , κ_R , or ρ_R/ρ_b from Sl geometry or projection structure.

14 What Opus III does not claim

Opus III does not claim:

1. a completed derivation of the Standard Model particle spectrum;
2. a completed derivation of $\eta_B \approx 6 \times 10^{-10}$;
3. a completed derivation of $\Omega_{\text{DM}}/\Omega_b \approx 5.3$;

4. a replacement for local QED, QCD, or electroweak reaction physics;
5. that positrons are forbidden in branch A ;
6. that branch B is an observable antimatter domain inside our universe;
7. that stable visible domain walls are produced;
8. that arbitrary Kibble defects automatically explain dark matter;
9. that the 40 MeV scale is a present-day crossing barrier;
10. that the neutrino sector is already solved.

The note is a framework paper. Its purpose is to define the corrected mechanism sharply enough that future work can either compute the missing quantities or fail.

15 Falsifiability and failure targets

A useful theory must be able to fail. The SI program fails if:

1. the universal orientation label χ cannot be defined independently of already-existing particle species;
2. the projection modes cannot reproduce Standard Model gauge representations even schematically;
3. gauge neutrality cannot be preserved in each branch-visible sector;
4. local QED requires branch-crossing in contradiction with the projection rule;
5. T_c has no calculable role in η_A , κ_R , or relic yield;
6. the relic sector cannot be made Standard Model neutral;
7. the relic number density n_R cannot be derived;
8. $\Omega_{\text{DM}}/\Omega_b$ is fitted independently from η_B ;
9. the relic sector reduces to ordinary asymmetric dark matter with no added projection prediction;
10. the neutrino sector contradicts the required orientation bookkeeping.

The strongest near-term target is:

$$\text{derive } \kappa_R = \frac{n_R}{n_b}. \quad (38)$$

Once κ_R is known, the relic mass follows:

$$m_R \approx \frac{5.3}{\kappa_R} m_p. \quad (39)$$

This is the cleanest bridge from abstract SI geometry to an observable dark matter scale.

16 Roadmap to Opus IV

Opus IV should not merely restate the philosophy. It should attempt one concrete calculation or no-go result.

The most useful routes are:

1. Projection-statistics route. Derive $\kappa_R = n_R/n_b$ from the number of visible matter modes versus neutral relic modes.
2. Index-operator route. Define a SI operator D_{SI} whose eigenmodes split into visible matter, conjugate branch, and neutral relic classes. Compute an index quotient.
3. Timing route. Show how $T_c \approx 40 \text{ MeV}$ enters η_A and κ_R through annihilation rate, expansion rate, and SI condensation.
4. Relic-mass route. Assume or derive κ_R and compute the implied m_R . Then compare with structure formation, self-interaction, and direct-detection constraints.
5. Neutrino route. Write the χ map for Dirac neutrinos. Then test whether a Majorana-compatible version exists. If not, state that confirmed Majorana neutrinos falsify the minimal model.
6. No-go route. Prove that no SI projection can simultaneously satisfy gauge neutrality, local QED, η_B , $\Omega_{\text{DM}}/\Omega_b$, and relic stability.

The no-go route is not a defeat. It would be a useful result.

17 Conclusion

Opus III reformulates polarized SI-membrane cosmology around one central inversion:

Particles are not the primary carriers of matter-antimatter asymmetry.

A universal matter-orientation label is primary, and particles are the quantized projection modes through which that label appears.

The observed universe is therefore not merely matter-dominated. It is a matter-oriented, gauge-neutral branch projection of a deeper structure. The conjugate orientation is assigned to a dynamically isolated branch, while neutral residual stress-energy of the SI transition may appear as a dark relic sector.

The scale near 40 MeV is not a particle barrier. It is the proposed condensation epoch at which the SI order parameter locks the universal orientation label into branch-classical populations during the annihilation/projection history.

The dark matter problem becomes a relic-statistics problem:

$$m_R n_R \approx 5.3 m_p n_b. \quad (40)$$

Thus the model must predict n_R/n_b . If it cannot, the relic sector is only a name. If it can, then the dark matter mass scale becomes an output rather than a fitted parameter.

The series therefore now has a clear burden:

Opus IV must derive a number.

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